Abstract

In many households and workplaces, fans are often left running unnecessarily, leading to wasted energy and increased electricity costs. Additionally, manually adjusting fan speed to match changing room temperatures can be inconvenient. There is a need for an automated system that can detect human presence and adjust fan speed dynamically based on room temperature and ensuring comfort without manual intervention. This project is designed to optimize fan operation based on room occupancy and temperature conditions. The system integrates an IR sensor to detect human presence and DHT11 temperature sensor to monitor ambient temperature. Using these inputs, the fan speed is automatically adjusted, ensuring energy-efficient operation and enhanced comfort. When the IR sensor detects a person's presence, the system activates and adjusts the fan speed proportionally to the temperature detected by the DHT11 sensor. The speed control is achieved through Pulse Width Modulation (PWM), ensuring smooth and precise control of the fan motor. If no presence is detected, the fan turns off, conserving energy. This project demonstrates the practical application of sensors and microcontroller-based automation for everyday use.

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List of Acronyms and Abbreviations

ADC	Analog to Digital Converter
CPU	Central Processing Unit
DC	Direct Current
DHT	Digital Temperature and Humidity Sensor
IDE	Integrated Development Environment
IR	Infrared Sensor
LCD	Liquid Crystal Display
PWM	Pulse Width Modulation
USB	Universal Serial Bus

Chapter 1

Introduction

With the advancement in technology, intelligent systems are introduced every day. Everything is getting more sophisticated and intelligible. There is an increase in the demand of cutting edge technology and smart electronic systems. Microcontrollers play a very important role in the development of the smart systems as brain is given to the system. Microcontrollers have become the heart of the new technologies that are being introduced daily. A microcontroller is mainly a single chip microprocessor suited for control and automation of machines and processes. Today, microcontrollers are used in many disciplines of life for carrying out automated tasks in a more accurate manner. Almost every modern day device including air conditioners, power tools, toys, office machines employ microcontrollers for their operation. Microcontroller essentially consists of Central Processing Unit (CPU), timers and counters, interrupts, memory, input/output ports, analog to digital converters (ADC) on a single chip. With this single chip integrated circuit design of the microcontroller the size of control board is reduced and power consumption is low.

An automated fan speed controller using a temperature sensor and an infrared (IR) sensor is an advanced system designed to regulate the speed of a fan based on environmental conditions and user presence. This innovative system integrates sensor technology and control mechanisms to improve energy efficiency and user convenience. The temperature sensor, typically a thermistor or a digital temperature module, continuously monitors the ambient temperature and adjusts the fan speed accordingly. For instance, at higher temperatures, the fan operates at a higher speed to provide optimal cooling, while at lower temperatures, it runs slower to conserve energy. The IR sensor adds an additional layer of functionality by detecting the presence of individuals in the room. If no one is detected within a certain range, the fan can automatically switch off or operate at a minimal speed, further enhancing energy efficiency. This system is particularly useful in modern homes, offices, and industrial settings where intelligent control of appliances is desired to minimize power consumption and reduce manual intervention. By combining these two sensors, the automated fan speed controller offers a smart, user-friendly solution to maintain comfort while optimizing energy usage.

Chapter 2

Literature Review

2.1 General Introduction

This section examines existing research, theories, and methodologies relevant to the project. It identifies knowledge gaps, establishes the context, and highlights advancements in the field. By analyzing previous studies, the review provides a foundation for the project, ensuring alignment with established practices while addressing unresolved challenges.

2.2 Literature Survey

1. Automatic fan control using temperature and room occupancy by B Kommey, JK Dunyo, BK Akowuah, Elvis Tamakloe, Mar-31, 2022 “Journal of Information Technology and Computer Engineering”. It focuses on the development of an automated system that controls the speed of a fan based on room temperature and occupancy.

2. An IOT Based Smart Fan Module by Anup Bind, Kiran Ashtankar, Mahesh Chahare, April 2020, “International Journal of Innovative Research in Technology”. The paper describes an automated fan speed controller using a DHT11 temperature sensor to adjust fan speed based on room temperature, while also utilizing a presence detection mechanism to turn the fan on or off, enhancing energy efficiency.

3. Design and Implementation of Automatic Room Temperature Controlled Fan using Arduino Uno and DHT11 Heat Sensor by Nur Afiqah Junizan, Amirrudin Abdul Razak, Bohendiran Balakrishnan, W.A.F.W. Othman “International Journal of Engineering Creativity and Innovation”. The proposed article describes the design of an automatic fan speed control system using microcontrollers and how temperature sensors like DHT11 can monitor ambient temperature and use PWM techniques to adjust the fan speed dynamically.

4. Automatic control system of fan by Liang Cui, 12 Jun 2013. The paper describes an automatic fan controller using a temperature sensor and an infrared sensor, enabling stepless speed adjustment based on environmental temperature changes, automatic fan activation at high temperatures, and deactivation at low temperatures, optimizing energy use.

2.3 Summary

The literature review examines studies on automatic fan speed controller using a temperature sensor and IR sensor explores the integration of sensors and microcontrollers to create energy- efficient systems. Studies highlight the use of microcontrollers like Arduino for seamless automation and control. This technology is widely applied in smart homes and industrial automation for enhanced efficiency and user convenience.

Chapter 3

Problem Statement and Objectives

3.1 Problem Statement

In conventional fan systems, the speed control is typically adjusted manually, which can be inconvenient and inefficient. In scenarios where the environment's temperature fluctuates frequently, manual adjustments can lead to discomfort, energy wastage, and unnecessary wear on the fan's components. Moreover, fans often remain operational even in the absence of individuals, further increasing energy consumption.

3.2 Objectives

1. To design and develop a system that automatically turns on the fan when a person is detected in the room and turns off the fan when the room is unoccupied.
2. To minimize energy consumption by operating the fan at appropriate speeds based on environmental conditions, reducing unnecessary power usage.

Chapter 4

Methodology and Implementation

4.1 Methodology

The system is an automated fan speed controller in which there are to sensors used, the hardware components are connected appropriately. The IR sensor is connected to digital pin 2 of the Arduino UNO to detect remote control signals, while the DHT11 temperature sensor is connected to digital pin 3 for measuring temperature and humidity. The DC fan is connected to a fan speed controller, such as an L293D driver, which is further interfaced with digital pin 9 of the Arduino for PWM-based speed control. The system is powered using a suitable power supply, ensuring compatibility with the components. Once the components are connected they can be controlled and monitored by software Arduino IDE, the code is executed in the software Arduino IDE, After uploading the code to the Arduino, the system is tested by observing the fan's behaviour in response to temperature changes and IR detection, adjusted the temperature thresholds in the code to customize the fan's activation range.

4.1 Block Diagram of the Project

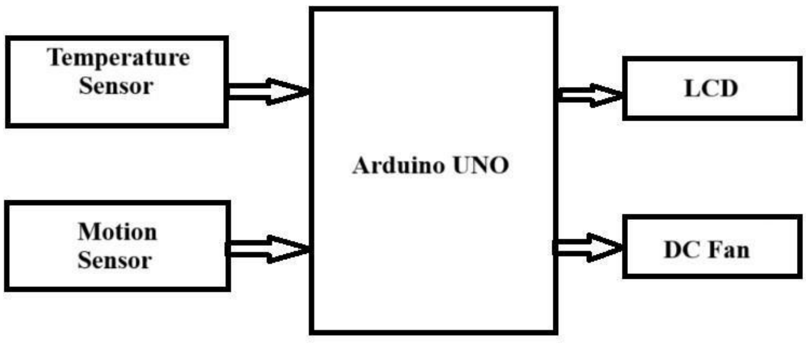

Fig 4.1: Block diagram of Automated fan speed controller

4.2 Flowchart of the Project

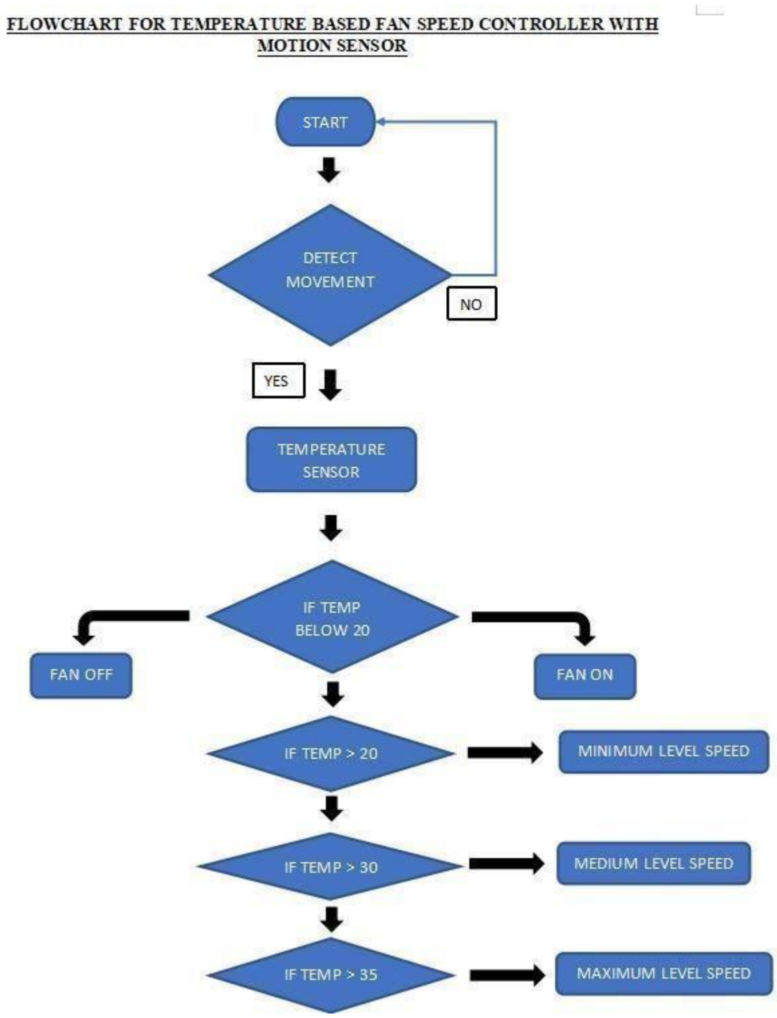

Fig 4.2: Flowchart of Automated fan speed controller

In the flowchart above, the process of this project is explained. First, there will be a detect movement section by the motion sensor. By the movement detection the sensor will light up. Then, there will be a temperature sensor section. If this temperature sensor detects the temperature below 20 the fan will turn off and if the temperature is above 20 then the fan will be working as given in the flowchart.

4.3 Hardware Requirements

1. Arduino UNO:-

The ATmega328-based Arduino Uno is a microcontroller board (data sheet). It has 14 digital I/O pins (six of which are PWM outputs), 6 analogue inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header, and a reset button. It comes with everything you need to support the microcontroller; simply connect it to a computer via USB or power it via an AC-to-DC adapter or battery to get started. The word "uno" means "one" in Italian and was chosen to mark a major redesign of the Arduino hardware and software. The ATmega328 on the board comes preprogrammed with a bootloader that allows uploading new code to it without the use of an external hardware programmer.

Fig 4.3: Arduino UNO Board

2. DHT11 Sensor:-

A popular temperature and humidity sensor is the DHT11. The sensor includes a dedicated NTC for temperature measurement and an 8-bit microcontroller for serial data output of temperature and humidity readings. Additionally factory calibrated, the sensor makes it simple to connect with other microcontrollers. The sensor can measure temperature from 0°C to 50°C and humidity from 20% to 90% with an accuracy of ±1°C and ±1%. The DHT11 sensor is used for many applications like to measure temperature and humidity, local weather station, automatic climate control and environment monitoring.

Fig 4.4: DHT11 Sensor

3. LCD Display:-

LCD (Liquid Crystal Display) screens are electronic display modules that have many applications. A 16x2 LCD display is a very basic module that is widely used in a variety of devices and circuits. The ASCII value of the character to be displayed on the LCD is represented by the data. The LCD 16×2 working principle is, it blocks the light rather than dissipate. The main benefits of using this module are inexpensive; simply programmable, animations, and there are no limitations for displaying custom characters, special and even animations, etc.,.

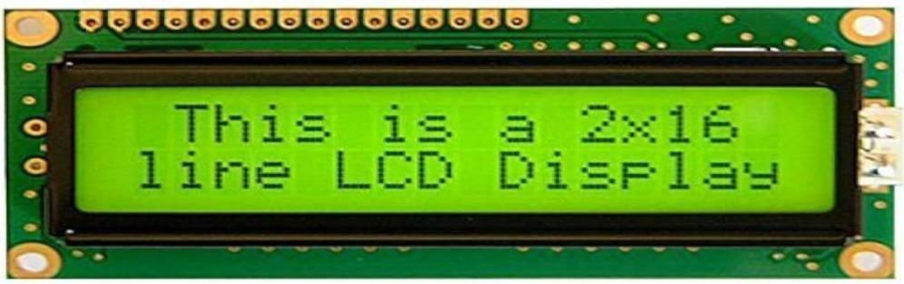

Fig 4.5: LCD Display

4. Fan: -

This component is controlled by the Atmega328p based on the temperature readings. It can be used to cool down the system if the temperature exceeds a predefined threshold. A **fan** is a powered machine that creates airflow. A fan consists of rotating vanes or blades, generally made of wood, plastic, or metal, which act on the air. The rotating assembly of blades and hub is known as impeller, rotor, or runner. Usually, it is contained within some form of housing or case.

Fig 4.6: Fan

5. IR Sensor:-

IR sensors are used for motion detection, proximity sensing, and temperature measurement in various applications. An infrared sensor (IR sensor) is a radiation-sensitive optoelectronic component with a spectral sensitivity in the infrared wavelength range 780 nm-50 µm. IR sensors are now widely used in motion detectors, which are used in building services to switch on lamps or in alarm systems to detect unwelcome guests. In a defined angle range, the sensor elements detect the heat radiation (infrared radiation) that changes over time and space due to the movement of people. Such infrared sensors only have to meet relatively low requirements and are low-cost mass-produced items.

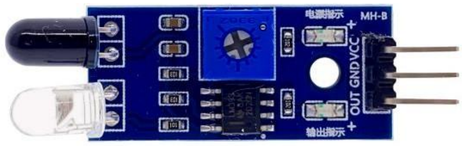

Fig 4.7: IR Sensor

6. Motor Driver Module:-

The L298N motor driver module is a widely used dual H-bridge motor driver that allows you to control the speed and direction of two DC motors. In this project, it is used to control the speed of the fan by regulating the voltage using a PWM signal. The L298N Motor Driver Module is a high power motor driver module for driving DC and Stepper Motors. This module consists of an L298 motor driver IC and a 78M05 5V regulator. L298N Module can control up to 4 DC motors, or 2 DC motors with directional and speed control.

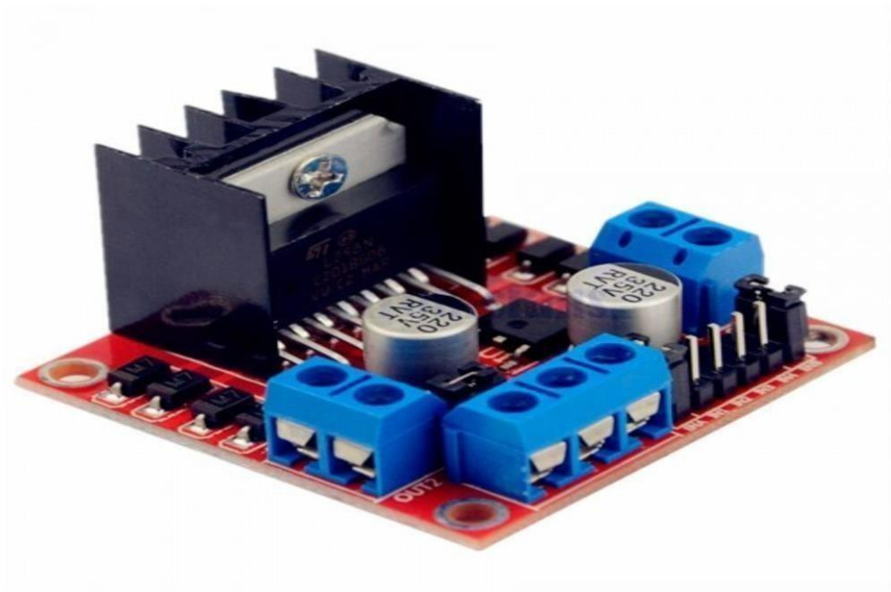

Fig 4.8: Motor Driver Module

7. **Breadboard:-**

A Breadboard, solderless breadboard or protoboard is a construction base used to build semi-permanent prototypes of electronic circuits. Unlike a perfboard or stripboard, breadboards do not require soldering or destruction of tracks and are reusable. Compared to more permanent circuit connection methods, modern breadboards have high parasitic capacitance, relatively high resistance, and less reliable connections, which are subject to jostle and physical degradation.

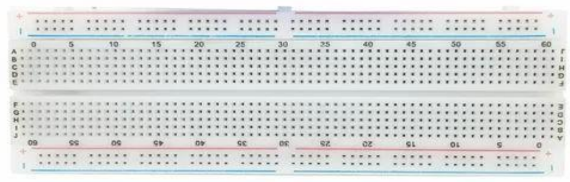

Fig 4.9: Breadboard

8. **Jumper wires:-**

Jumper wires are simply wires that have connector pins at each end, allowing them to be used to connect two points to each other without soldering. Jumper wires are typically used with breadboards and other prototyping tools in order to make it easy to change a circuit as needed. There are three types of Jumper wires:- male-to-male, male-to-female and female-to-female.

Fig 4.10: Jumper wires

4.4 Software Requirements

Arduino IDE

The Arduino IDE is an open-source software, which is used to write and upload code to the Arduino boards. The IDE application is suitable for different operating systems such as Windows, Mac OS X, and Linux. It supports the programming languages C and C++. IDE stands for Integrated Development Environment.

Fig 4.11: Arduino IDE

Chapter 6

Result and Discussions

6.1 Result Analysis

The result analysis of Automated fan speed controller using Temperature sensor and IR sensor. First, the IR sensor detected the motion and turned on the fan. Then the temperature readings from sensor such as DHT11 is read. The fan speed is adjusted according to the detected temperature. If the temperature is below 34°C, the fan remains off. If the temperature is above 38°C then the fan runs at maximum speed.

According to the observations, the surrounding environment had a temperature of 33 degrees Celsius. So, to run the fan we gave heat to the DHT11 sensor through the lighter. Then the temperature started increasing and the fan started to rotate. When the temperature was at 34°C, the fan was rotating at low speed. When the temperature was in between 34°C to 38°C, then the fan speed was at medium. When the temperature reached 38°C, then the fan speed was at maximum. Then the temperature decreased after few minutes, and the fan stopped rotating.

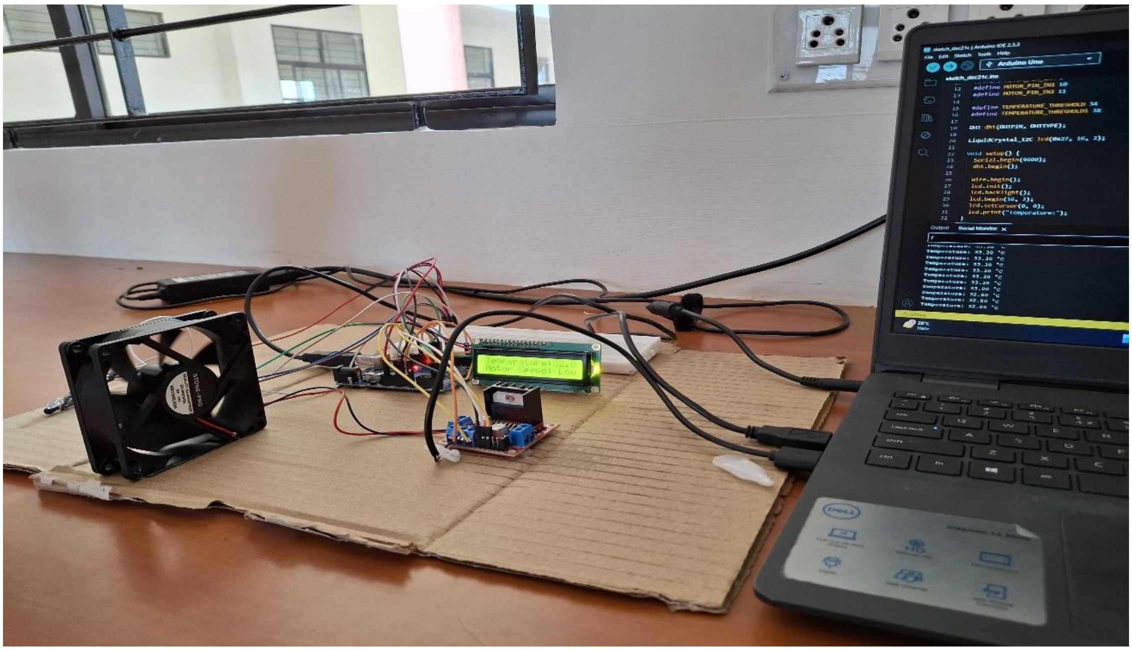

Fig 6.1: Model of the project

6.2 Summary

An automatic fan speed controller using a DHT11 sensor and an IR sensor can be implemented to enhance comfort and energy efficiency by adjusting the fan's speed based on environmental conditions and room occupancy. The DHT11 sensor measures temperature and humidity, allowing the system to regulate the fan speed according to predefined temperature thresholds. An IR sensor detects the presence of a person in the room, ensuring the fan operates only when someone is present. A microcontroller processes data from both sensors and controls the fan speed using Pulse Width Modulation (PWM) through a motor driver. This automation eliminates the need for manual adjustments, making it ideal for smart homes, offices, and energy-conscious systems. By reducing power consumption when the room is unoccupied or temperatures are moderate, this system provides a comfortable environment while promoting energy efficiency.

Conclusion

At the end of this project, we hope that this project called temperature based fan speed controller with motion sensor will contribute and benefit everyone, especially our country. This project has applications in both the home and the workplace. This will aid in energy or electricity conservation. To monitor environments that are not comfortable or possible for humans to monitor, especially for long periods of time. Prevents energy waste when it isn't hot enough for a fan to be needed.

REFERENCE

[1] B. Kommey, J. K. Dunyo, B. K. Akowuah, and E. Tamakloe, "Automatic fan control using temperature and room occupancy," Journal of Information Technology and Computer Engineering, vol. 4, no. 1, pp. 23–30, Mar. 2022.

[2] K. H. Danjuma, "Design and construction of an automatic temperature controlled fan", International Journal of Engineering and Technology, vol.7, no.2, 2018.

[3] M. K. H. Majumdar, H. Biswas, M. H. A. Shaim and K. T. Ahmmed, "Automated energy saving and safety system," 2014 International Conference on Electrical Engineering and Information & Communication Technology, 2014, pp. 1-6.

[4] S. Ragul, Y. S, V. Vijayabalan, B. Venkatasamy, L. Kalaivani and F. A. Jeffrey Vaz, "IoT based Smart Controller for Ceiling Fan," 2023 7th International Conference on Trends in Electronics and Informatics (ICOEI), 2023, pp. 376-382

[5] M. Saad, H. Abdoalgader, and M. Mohamed, "Automatic Fan Speed Control System Using Microcontroller," 6th Int'l Conference on Electrical, Electronics & Civil Engineering (ICEECE'2014), pp. 86-89, Nov. 27-28, 2014.

APPENDICES

```
#include <SoftwareSerial.h>
#include<Arduino.h>
#include
<Wire.h>
#include
<dht.h> dht
DHT;
SoftwareSerial ss(2, 3);
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 16, 2);
#define IR A1
#define ALARM 13
 #define FAN 9
#define
DHT11_PIN A2
int Mode=0;  float
Press; int
Alm1=0;  int
Alm2=0;  int
Alm3=0;  float
val=0;  int
tempPin = 1;
float Flame;  int
TIMER=40; float
SPD=0;  float
Smoke1=0;  int
Timerx=0; String
Status="STOP";
float
Hum,Temp,Sens1
;  float Sens2,WP;
float Speed=125;
```

```
int ST=0;  float
Smoke; void
setup(void)
{
pinMode(FAN,OUTPUT);
digitalWrite(FAN,LOW);
Serial.begin(9600);
ss.begin(9600);
pinMode(IR,INPUT);
pinMode(OUTPUT);
pinMode(ALARM,OUTP
UT);
 lcd.begin(); lcd.clear();
lcd.setCursor(0, 0);
lcd.print("WELCOME");
delay(3000);
digitalWrite(HIGH);
delay(40);
digitalWrite(LOW);
delay(40);
digitalWrite(HIGH);
delay(40);
digitalWrite(LOW);
delay(40); }
void loop(void) { int chk =
DHT.read11(DHT11_PIN); switch (chk)
{ case DHTLIB_OK:
Serial.print("OK,\t"); break; case DHTLIB_ERROR_CHECKSUM:
Serial.print("Checksum error,\t"); break; case DHTLIB_ERROR_TIMEOUT:
Serial.print("Time out error,\t"); break; case
DHTLIB_ERROR_CONNECT:  Serial.print("Connect error,\t");
break; case DHTLIB_ERROR_ACK_L:
Serial.print("Ack Low error,\t"); break; case
DHTLIB_ERROR_ACK_H: //Serial.print("Ack High error,\t");
break; default:
```

```
//Serial.print("Unknown error,\t"); break;
}
Temp=DHT.temperature ; Hum=DHT.humidity;
lcd.clear();
lcd.setCursor(0,
0);
lcd.print("T:");
lcd.print(Temp,1);
lcd.print("c");
lcd.print(" H:");
lcd.print(Hum,1);
l cd.print("%"); if
(digitalRead(IR)=
=1){
lcd.setCursor(0,
1);
lcd.print("MOTION
DETECTED!"); if (ST==0){
ST=1;
digitalWrite(FAN,HIGH);
delay(100);
digitalWrite(FAN,LOW);
}
TIMER=20;
} if (TIMER>0){
TIMER--;
Serial.println(TIMER); if
(TIMER==0){ if
(ST==1){
digitalWrite(FAN,HIGH);
delay(100);
digitalWrite(FAN,LOW);
delay(100); d
igitalWrite(FAN,HIGH);
delay(100);
```

```
digitalWrite(FAN,LOW);
delay(100);
digitalWrite(FAN,HIGH);
delay(100);
digitalWrite(FAN,LOW);
ST=0; }
} } if
(digitalRead(IR)==1
){ } delay(800); }
```